Pressure Formula Triangle

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS The triangle links pressure (P), force (F), and area (A). Cover the unknown to read the formula: $P = F \div A$, $F = P \times A$, and $A = F \div P$.

Round 1 — Retrieve It

1. What three quantities does the pressure triangle connect?
2. Write the formula for pressure.
3. Write the formula for force.
4. Which variable do you cover to find area, and what formula results?

Round 2 — Sketch It

Draw the pressure triangle. Put pressure on top; put force and area on the bottom. Circle the two quantities you would multiply to find force.

Pressure — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. A force of 20 N presses on an area of 4 m^2. What is the pressure?
2. A pressure of 50 Pa acts over an area of 3 m^2. What force is applied?
3. A force of 240 N produces a pressure of 6 Pa. Over what area does it act?

Round 4 — Reason from Evidence

The same 60 N force presses on two surfaces: one has an area of 2 m^2, the other 6 m^2. Use the triangle to find the pressure on each, then say which surface feels more pressure and why a sharp blade or a snowshoe works the way it does.

Pressure — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain how the pressure triangle helps you find a missing quantity. Use one example that is not printed on the cue card.

C

Connect

Connect pressure to real examples. Explain why a sharp knife (small area) cuts easily as high pressure, and why snowshoes (large area) keep you on top of the snow as low pressure. Tie each to $P = F \div A$.

E

Extend

Create your own problem that asks for area. Give the numbers and show the work using $A = F \div P$.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Pressure

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Pressure, force, and area.
2. $P = F \div A$.
3. $F = P \times A$.
4. Cover A; $A = F \div P$.

Sketch It — look for

Pressure on top, with force and area on the bottom. Force is found by multiplying pressure × area (the two bottom quantities).

Apply It

1. $P = F \div A = 20 \div 4 =$ **5 Pa**.
2. $F = P \times A = 50 \times 3 =$ **150 N**.
3. $A = F \div P = 240 \div 6 =$ **40 m^2**.

Reason from Evidence — look for

Small area: $P = 60 \div 2 = 30$ Pa. Large area: $P = 60 \div 6 = 10$ Pa. The 2 m^2 surface feels more pressure because the same force is concentrated over a smaller area. A sharp blade puts force on a tiny area for high pressure to cut; a snowshoe spreads force over a large area for low pressure to stay on top of the snow.

ACE — Extend look-for

Many answers possible. Check that the problem asks for area and the work correctly uses $A = F \div P$.

> **Suggested use.** Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.